SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA10 / CSA1014	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	Apply advanced methodologies and agile project management techniques to manage software projects effectively		
Student Name:	Manish Kumar L	**Reg. No.:**	1924260185
Date:	05/10/2026	**Duration:**	60 minutes

Code of Conduct

*I **Manish KUmar L / 1924260185** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: Manish Kumar

Annexure A – Case Study Analysis

Instructions to Students:

Each student will be allotted one problem statement. Analyze the assigned problem systematically by following the steps given below. Your report should demonstrate your understanding of software engineering concepts and your ability to design an appropriate software solution.

Based on your allotted problem statement, prepare a report by answering the following questions:

1. Understand the Problem Statement

- Explain the problem in your own words.
- Identify the objectives of the proposed system.
- State the scope of the problem.

2. Identify Key Stakeholders and Challenges

- Identify all the stakeholders involved in the system.
- Explain the roles and responsibilities of each stakeholder.
- Discuss the major challenges and issues faced by the stakeholders.

3. Analyze the Current Approach

- Describe the existing system or current process.
- Identify its strengths and limitations.
- Analyze the problems or gaps that need to be addressed.

4. Propose a Suitable Solution Using Software Engineering Principles

- Design a software-based solution for the given problem.
- Describe the major functional modules of the proposed system.
- Identify the functional and non-functional requirements.
- Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.

5. Justify the Chosen Methodology/Framework

- Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).
- Justify why the selected methodology/framework is suitable for the proposed system.
- Explain how it supports successful project development and maintenance.

6. Discuss Expected Outcomes and Limitations

- Describe the expected benefits of the proposed solution.
- Explain how the solution addresses the identified challenges.
- Discuss potential risks, implementation challenges, and limitations.
- Suggest possible future enhancements.

Question

Intelligent Digital Twin Monitoring Platform for Industrial Assets

Industry Context

Industrial organizations need continuous visibility into complex assets such as turbines, pumps, compressors, and production systems.

Problem Statement

Design and develop a digital-twin platform that maintains virtual representations of physical assets and synchronizes operational, maintenance, sensor, and performance information.

Objectives

- Create digital asset models.
- Synchronize sensor data.
- Monitor equipment status.
- Maintain maintenance history.
- Visualize asset performance.
- Identify abnormal conditions.

Expected Outcomes

- Improved asset visibility.
- Better maintenance planning.
- Faster fault identification.
- Reduced downtime.
- Improved operational decision-making.

1. Understand the Problem Statement

The proposed system addresses the difficulty industrial organizations face in obtaining a unified, continuously updated view of complex physical assets. Turbines, pumps, compressors, and production systems generate large amounts of operational, sensor, maintenance, and performance information. When this information is separated across different systems, it becomes difficult to understand the current condition of an asset, trace its maintenance history, and identify abnormal behavior early.

The solution is an Intelligent Digital Twin Monitoring Platform. It creates a virtual representation of each physical asset and keeps the representation synchronized with relevant real-world information. The platform provides a common view of asset status, sensor readings, maintenance records, performance indicators, and abnormal conditions.

Objectives:

- Create and maintain digital models of industrial assets.
- Synchronize sensor and operational data with the corresponding digital twin.
- Provide continuous monitoring of equipment status and performance.

- Maintain a centralized and traceable maintenance history.
- Visualize asset condition and performance through dashboards and reports.
- Identify abnormal conditions so that responsible personnel can investigate them quickly.

Scope: The platform covers asset registration and modeling, sensor-data ingestion, digital-twin synchronization, real-time monitoring, maintenance-history management, performance visualization, abnormal-condition detection, alerts, reporting, and user access control. The scope focuses on software support for monitoring and decision-making; physical repair or automatic control of machinery is outside the stated scope unless added as a future enhancement.

2. Identify Key Stakeholders and Challenges

Key stakeholders and their roles are:

- Plant/Operations Managers – monitor overall equipment availability, production impact, and operational performance and use dashboards for decision-making.
- Maintenance Engineers and Technicians – review equipment condition, maintenance history, alerts, and recommended inspection or maintenance actions.
- Reliability Engineers – analyze performance trends, recurring abnormalities, and asset health indicators to improve reliability.
- Control Room/Operations Personnel – observe current equipment status and respond to abnormal-condition alerts.
- Data/IoT Engineers – connect sensors and data sources, maintain data pipelines, and ensure reliable synchronization.
- Software/System Administrators – manage users, permissions, platform configuration, integrations, backups, and system availability.
- Management/Decision Makers – use summarized performance information to support maintenance planning, resource allocation, and operational decisions.

Major challenges include high-volume and continuous sensor data, heterogeneous equipment and data sources, data-quality problems, synchronization delays, cybersecurity requirements, integration with existing industrial systems, false or excessive alerts, scalability across many assets, and the need for clear visualizations that can be understood quickly by operational users.

3. Analyze the Current Approach

A typical current approach may involve separate monitoring systems, sensor interfaces, maintenance records, spreadsheets or databases, and operator observations. Equipment information may therefore be distributed across multiple sources. Staff may need to compare current readings with historical maintenance information manually before determining whether an abnormal condition is significant.

Strengths of the current approach include the availability of existing sensor infrastructure, established maintenance procedures, experienced personnel, and historical records that can support troubleshooting.

Limitations include fragmented information, limited real-time synchronization between operational and maintenance data, difficulty obtaining a complete asset view, manual analysis of historical information, delayed

identification of abnormal conditions, inconsistent data formats, and limited ability to compare live performance with historical behavior.

The main gap is the absence of a unified digital representation that connects the physical asset to its current sensor state, operational performance, and maintenance history. The proposed platform closes this gap by bringing these information sources together around a digital twin.

4. Propose a Suitable Solution Using Software Engineering Principles

The proposed solution is a modular web-based Intelligent Digital Twin Monitoring Platform connected to industrial data sources. Each physical asset is represented by a digital twin containing identity, specifications, current state, sensor mappings, performance indicators, maintenance history, and abnormal-condition information.

Major functional modules:

- Asset & Digital Twin Management – create assets, define asset types, configure attributes, relationships, sensor mappings, and digital-twin profiles.
- Sensor/Data Ingestion – collect readings from connected sensors and industrial data sources, validate incoming data, normalize formats, timestamp records, and route data to the appropriate asset.
- Real-Time Monitoring – display current equipment status, sensor values, operating states, and health indicators.
- Maintenance Management – record inspections, faults, work orders, maintenance activities, dates, technicians, observations, and outcomes.
- Performance Visualization – provide dashboards, trends, KPIs, historical comparisons, and asset-level performance views.
- Abnormal Condition Detection – apply configured thresholds, rules, and trend checks to identify unusual conditions and generate alerts.
- Alert & Notification Management – prioritize alerts and notify authorized personnel according to configured rules.
- Reporting & Analytics – generate asset, maintenance, performance, and abnormal-condition reports for operational and management use.
- User & Security Management – provide authentication, role-based access control, audit logging, and secure configuration.

Functional requirements:

- The system shall create, update, view, and deactivate digital asset models.
- The system shall associate sensor streams with the correct digital asset.
- The system shall receive, validate, store, and display sensor information.
- The system shall show current equipment status and historical trends.
- The system shall record and retrieve maintenance history.
- The system shall identify configured abnormal conditions and generate alerts.
- The system shall provide dashboards, reports, and search/filter functions.
- The system shall restrict functions and information according to user roles.

Non-functional requirements:

- Reliability – monitoring and data services should remain available and recover from failures without unnecessary data loss.
- Scalability – the platform should support increasing numbers of assets, sensors, users, and data points.
- Performance – current asset information and alerts should be presented with low operational delay.
- Security – authentication, authorization, encryption, audit logs, and secure integrations should protect industrial information.
- Maintainability – modular components, clear interfaces, documentation, logging, and automated testing should make changes easier to implement.
- Usability – dashboards should present important asset information clearly for different user roles.
- Data integrity – timestamps, validation, duplicate handling, and consistent asset identifiers should preserve trustworthy records.

Software engineering principles are incorporated through modularity (separate services/modules for assets, data, monitoring, maintenance, alerts, and reporting), scalability (loosely coupled services and scalable data storage), maintainability (well-defined interfaces, documentation, testing, and logging), reliability (fault handling and recovery), usability (role-specific dashboards), and security (least-privilege access, secure communication, and audit trails).

5. Justify the Chosen Methodology/Framework

Agile with Scrum is the most suitable development framework for this platform. A digital-twin monitoring system has several user groups and integrations, and requirements can become clearer as stakeholders see working dashboards, asset models, and monitoring features. Scrum supports iterative development through short sprints, frequent demonstrations, stakeholder feedback, and continuous prioritization.

The project can be organized into incremental releases. An early sprint can establish the asset model and user management; later sprints can add sensor synchronization, monitoring dashboards, maintenance history, abnormal-condition detection, and reporting. Integration and security testing can be performed continuously rather than being postponed until the end.

Agile also supports successful maintenance because the platform can evolve as new asset types, sensors, analytics rules, and operational requirements are introduced. A prioritized product backlog allows the team to handle changing requirements while maintaining a working product at the end of each sprint.

6. Discuss Expected Outcomes and Limitations

Expected benefits:

- Improved asset visibility through a unified digital representation of physical equipment.
- Better maintenance planning using current condition information and historical maintenance records.
- Faster fault identification through continuous monitoring and abnormal-condition alerts.
- Reduced downtime by enabling earlier investigation and more informed maintenance decisions.
- Improved operational decision-making through dashboards, trends, KPIs, and consolidated asset information.

The solution addresses the identified challenges by integrating data sources, connecting sensor information to digital assets, centralizing maintenance history, providing continuous visualization, and applying rules or analytics to highlight abnormal conditions. This reduces the need for manual comparison of disconnected information.

Potential risks and implementation challenges include unreliable or missing sensor data, inaccurate asset-to-sensor mapping, integration difficulties with legacy industrial systems, network interruptions, cybersecurity threats, storage and processing demands from high-frequency sensor streams, false alarms, and user adoption challenges. The accuracy of abnormal-condition detection also depends on the quality of thresholds, rules, historical data, and models used.

Limitations: The platform is primarily a monitoring and decision-support solution. It does not by itself guarantee accurate diagnosis of every equipment fault, eliminate the need for skilled personnel, or physically repair equipment. Digital-twin quality also depends on the accuracy and timeliness of the data synchronized from physical assets.

Future enhancements may include predictive maintenance models, machine-learning-based anomaly detection, remaining-useful-life estimation, simulation capabilities, mobile monitoring, automated work-order recommendations, integration with enterprise maintenance systems, and advanced digital-twin analytics for scenario testing and optimization.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding (30%)	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis (25%)	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis (20%)	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.
Proposed Software Solution & Software Engineering Principles (15%)	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability, maintainability, security, etc.).	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.
Methodology/ Framework Justification (10%)	Selects the most suitable SDLC/model/framework and provides strong technical justification.	Suitable methodology selected with adequate justification.	Methodology selected but justification is weak.	Outcomes, limitations, or future enhancements are missing; report lacks organization and clarity.

Criteria	**Mark**
Problem Understanding (30%)	/5
Stakeholder & Challenge Analysis (25%)	/5

Current System Analysis (20%)	/5
Proposed Software Solution & Software Engineering Principles (15%)	/5
Methodology/ Framework Justification (10%)	/5
TOTAL	/25